

0 Instructions

Homework is not meant to be turned in. Please refer to <https://courses.grainger.illinois.edu/cs446/sp2026> for course policy on homeworks. Use this homework to help you study for quizzes and exams, but it is not a comprehensive or sufficient study tool.

1 Design Neural Networks for Simple Functions

In this problem you will be hand designing a simple neural network $f(x)$

$$f(x) = \mathbf{w}_1^\top \sigma(\mathbf{w}_0 x + \mathbf{b}_0)$$

to model specific functions. In each case you can assume $x \in \mathbb{R}$ and $\mathbf{w}_0 x$ is a vector-scalar product $\mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ (e.g., $[0, 1]^\top x = [0, x]^\top$). You will need to provide appropriate weights $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0 \in \mathbb{R}^d$ for some given d . In other words, each neural network has one input neuron, d hidden neurons, and one output neuron. The goal is to choose a specific setting of $\mathbf{w}_1, \mathbf{w}_2$ and $\mathbf{b}_0$ such that the neural network functions as desired.

1. Fix $d = 2$ and $\mathbf{b}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we want to find $\mathbf{w}_0, \mathbf{w}_1 \in \mathbb{R}^2$ such that

$$f(x) = \mathbf{w}_1^\top \sigma(\mathbf{w}_0 x) = x, \quad \forall x \in \mathbb{R}$$

$$\text{where } \sigma(x) = \text{ReLU}(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

2. Fix $d = 2$, we want to find $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0 \in \mathbb{R}^2$ such that

$$f(x) = \mathbf{w}_1^\top \sigma(\mathbf{w}_0 x + \mathbf{b}_0) = mx + b, \quad \forall x \in \mathbb{R}$$

where $\sigma(x) = \text{ReLU}(x)$.

3. Fix $d = 1$ and $b_0 = 0$, we want to find $w_0, w_1 \in \mathbb{R}$ such that

$$f(x) = w_1^\top \sigma(w_0 x) = b, \quad \forall x \in \mathbb{R}$$

where $\sigma(x) = \text{sigmoid}(x) = \frac{1}{1+e^{-x}}$

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4. Fix $d = 3$, we want to find $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0 \in \mathbb{R}^3$, $\sigma(x) = \text{ReLU}(x)$, and

$$f(x) = \mathbf{w}_1^\top \sigma(\mathbf{w}_0 x + \mathbf{b}_0) = \begin{cases} 0, & x \leq -2 \\ 3x + 6, & -2 \leq x \leq 0 \\ -3x + 6, & 0 \leq x \leq 2 \\ 0, & x \geq 2 \end{cases}$$

5. Do there exist $\mathbf{w}_0, \mathbf{w}_1 \in \mathbb{R}^d$ such that

$$f(x) = \mathbf{w}_1^\top \sigma(\mathbf{w}_0 x) = x^2, \quad \forall x \in \mathbb{R}$$

where $\sigma(x) = \text{ReLU}(x)$? Explain why or why not.

6. For any $\varepsilon > 0$, do there exist fixed finite dimensional $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0 \in \mathbb{R}^d$ such that

$$|f(x) - x^2| = |\mathbf{w}_1^\top \sigma(\mathbf{w}_0 x + \mathbf{b}_0) - x^2| < \varepsilon, \quad \forall x \in \mathbb{R}$$

where $\sigma(x) = \text{ReLU}(x)$? Explain why or why not. If yes describe a procedure to construct $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0$ at a high level.

7. For any $\varepsilon > 0$, do there exist fixed finite dimensional $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0$ such that

$$|f(x) - x^2| = |\mathbf{w}_1^\top \sigma(\mathbf{w}_0 x + \mathbf{b}_0) - x^2| < \varepsilon, \quad \forall x \in [0, 1]?$$

where $\sigma = \text{ReLU}$? Explain why or why not. If yes describe a procedure to construct $\mathbf{w}_0, \mathbf{w}_1, \mathbf{b}_0$ at a high level.

2 Backpropagation

2.1 Backpropagation in a Two-Layer Neural Network

Consider a two-layer fully connected neural network for binary classification.

The forward propagation is defined as:

$$\begin{aligned}
\mathbf{z}^{(1)} &= \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \\
\mathbf{a}^{(1)} &= \text{ReLU}(\mathbf{z}^{(1)}) \\
z^{(2)} &= \mathbf{W}^{(2)}\mathbf{a}^{(1)} + b^{(2)} \\
\hat{y} &= \sigma(z^{(2)})
\end{aligned}$$

The loss function is binary cross-entropy, where the groundtruth label $y \in \{0, 1\}$:

$$L(y, \hat{y}) = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y})).$$

Assume: $\mathbf{x} \in \mathbb{R}^3$, hidden layer has 4 neurons and output layer has 1 neuron. The sigmoid function is defined as $\sigma(z) = \frac{1}{1+e^{-z}}$.

1. Determine the shape (dimension) of the following quantities: $\mathbf{W}^{(1)}$, $\mathbf{b}^{(1)}$, $\mathbf{W}^{(2)}$ and $\frac{\partial L}{\partial \mathbf{W}^{(1)}}$.
2. Suppose we are given the following setting of variables:

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{W}^{(1)} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad \mathbf{b}^{(1)} = \mathbf{0}, \quad \mathbf{W}^{(2)} = [1 \quad -1 \quad 0 \quad 1], \quad b^{(2)} = 0, \quad y = 1$$

Compute the following expressions (you may keep the answer in terms of e without numerical approximation): $\mathbf{z}^{(1)}$, $\mathbf{a}^{(1)}$, $z^{(2)}$, $\frac{\partial L}{\partial \mathbf{W}^{(2)}}$ and $\frac{\partial L}{\partial \mathbf{W}^{(1)}}$.

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2.2 Backpropagation through time (BPTT)

1. Consider a recurrent model with recurrence

$$h_t = w(x_t + h_{t-1}), \quad h_0 = 0.$$

We assume $x_t, h_t, w \in \mathbb{R}$. Draw the computation graph obtained by unrolling this RNN for 3 timesteps. You should include 3 inputs x_1, x_2, x_3 , the outputs h_0, h_1, h_2, h_3 , and refer to intermediate values

$$y_t = x_t + h_{t-1}.$$

2. Compute the forward pass in the above model, i.e., compute $h_0, h_1, h_2, h_3, y_1, y_2, y_3$ as functions of w, x_1, x_2, x_3 .
3. Compute

$$\frac{\partial h_3}{\partial w}$$

in the above model using a backward pass and your values from the forward pass. **Hint:** You can directly take the derivative of h_3 from the previous question to check your work, but you must show your work for a proper backprop including intermediate partial derivatives.

4. For a sequence

$$\mathbf{z} = [x_1, \dots, x_T],$$

consider the recurrent model

$$f(\mathbf{z}) = h_T, \quad h_t = w \sigma(x_t + h_{t-1}), \quad h_0 = 0.$$

Derive

$$\frac{\partial f}{\partial h_1}, \quad \frac{\partial h_T}{\partial h_1}.$$

You may use σ' to represent the derivative of the nonlinearity σ .

5. Using your previous answer as justification, explain the exploding/vanishing gradient problem in RNNs.

3 Convolutional Neural Network (CNN)

3.1 Convolutional Layers

1. Consider an input feature map of size

$$3 \times 32 \times 32$$

where the dimensions are ordered as $C \times H \times W$ (C: channel, H: height, W: width).

A convolutional layer with:

- 16 filters,
- kernel size 3×3 ,
- stride 1,
- zero-padding 1

is applied.

- (a) What is the size of the output feature map (in $C \times H \times W$ format)?
 - (b) How many learnable parameters does this convolutional layer have (including bias terms)?
2. Consider the following single-channel input feature map:

$$X = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 3 & 1 \end{bmatrix}$$

A convolutional layer with the following kernel is applied:

$$K = \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{bmatrix}$$

The convolution uses:

- stride 2
- no padding
- no bias term

- (a) What is the size of the output feature map?

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- (b) Compute the full output feature map.
- (c) Does this convolution operation perform the same function as average pooling? Explain your reasoning.

3.2 Pooling Layers in CNN

1. Consider a 2D input feature map:

$$X = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Apply a 2×2 max pooling operation with stride 2. You can assume stride and padding is symmetric in both spatial dimensions. What is the resulting feature map without padding? What if zero-padding was applied with padding of 1?

2. Consider a 6×6 input feature map with a 3×3 max pooling layer with stride 1. What is the size of the output feature map?
3. For a 5×5 input matrix, if a 2×2 pooling operation with stride 2 is applied, what would be the output size if zero-padding was applied with a padding of 1?

3.3 Equivalent Linear Layer of a CNN

Consider a convolutional neural network consisting of a single convolutional layer with no bias and no activation function.

Let the input be a 3×3 grayscale image

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

The convolution layer has the following specifications:

- One convolution kernel
- Kernel size: 2×2
- Stride: 1
- No padding

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- No bias
- No activation function

The kernel is

$$K = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

The convolution produces an output feature map $Y \in \mathbb{R}^{2 \times 2}$.

Define the flattened input vector

$$\mathbf{x} = [x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}]^\top \in \mathbb{R}^9.$$

Define the flattened output vector

$$\mathbf{y} = [y_{11}, y_{12}, y_{21}, y_{22}]^\top \in \mathbb{R}^4.$$

Questions

- Write the explicit expressions of $y_{11}, y_{12}, y_{21}, y_{22}$ in terms of the input pixels and kernel parameters.
- Construct a matrix $\mathbf{W} \in \mathbb{R}^{4 \times 9}$ such that

$$\mathbf{y} = \mathbf{W}\mathbf{x}.$$

- Describe the structural properties of $\mathbf{W}$. Is it dense or sparse? What pattern does it exhibit?

4 Variational Autoencoder (VAE)

In this problem, we review the algorithmic pipeline of the Variational Autoencoder (VAE) and analyze its theoretical properties.

Consider the following generative model:

$$z \sim p(z) = \mathcal{N}(0, I), \quad (1)$$

$$x \sim p_\theta(x | z), \quad (2)$$

where z is a latent variable and x is the observed data. The marginal likelihood of the data is given by

$$p_\theta(x) = \int p_\theta(x | z)p(z) dz, \quad (3)$$

which is generally intractable.

To enable efficient learning, VAE introduces an approximate posterior

$$q_\phi(z | x), \quad (4)$$

parameterized by an encoder network, and maximizes the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x | z)] - D_{KL}(q_\phi(z | x) \| p(z)). \quad (5)$$

Assume the encoder outputs a diagonal Gaussian:

$$q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x))). \quad (6)$$

During training, sampling is implemented using the reparameterization trick:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

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Answer the following questions.

1. **Approximations in VAE.**

Identify and clearly explain all sources of approximation involved in VAE training. Your answer should refer to the probabilistic formulation above.

2. **Closed-form KL Divergence.**

Assume

$$q_\phi(z \mid x) = \mathcal{N}(\mu, \sigma^2) \quad \text{and} \quad p(z) = \mathcal{N}(0, 1)$$

Derive the closed-form expression of

$$D_{KL}(q_\phi(z \mid x) \parallel p(z)) .$$

Show all key steps in your derivation.

3. **Smoothness of VAE Generations.**

VAE-generated samples are often observed to vary smoothly when interpolating in the latent space.

Explain, from the perspective of the ELBO objective, why VAE tends to produce smooth latent representations and smooth generations.